
Datasets
① Column Profiler
② Ring Builder
③ Budgeted Verifier
Join Graph
datasets/
$\mathcal{D}_1$
erp
project
ledger
timesheet
+ more tables
$\mathcal{D}_2$
$\mathcal{D}_3$
$\mathcal{D}_n$
Value Sketches
MinHash, HLL
Uniqueness, Type, Value Pattern
Format Normalization
Name + Description Embedding
Key Candidates
$u(c) \geq \tau_u$
Candidate Index
LSH + ANN
Prune generic and type-incompatible columns
R_1
R_2
R_k
one ring per key domain: • owner ○ member
Cost-based Optimizer
thresholds, batches
Distilled Model
accept / reject
uncertain
uncertain
labels
LLM
Ring Prompt
owner, members, uncertain
Transitive Propagation
over confirmed rings
star per key domain
$G(\mathcal{R})$